return the SHA-256 hash of the session cookie as the token ID, rather than the real value. Java's `MessageDigest` class (in the `java.security` package) implements a number of cryptographic hash functions, and SHA-256 is implemented by all current Java environments. As SHA-256 returns a byte array and the token ID should be a `String`, you can Base64-encode the result to generate a string that is safe to store in a cookie or header. It is common to use the URL-safe variant of Base64 in web APIs, as it can be used almost anywhere in a HTTP request without additional encoding, so that is what you will use here. Listing 4.10 shows a simplified interface to the standard Java Base64 encoding and decoding libraries implementing the URL-safe variant. Create a new file named Base64url.java inside the src/main/java/com/manning/apisecurityinaction/token folder with the contents of the listing.

Listing 4.10 URL-safe Base64 encoding

```
package com.manning.apisecurityinaction.token;

import java.util.Base64;

public class Base64url {
    private static final Base64.Encoder encoder =    #A
            Base64.getUrlEncoder().withoutPadding();    #A
    private static final Base64.Decoder decoder =    #A
            Base64.getUrlDecoder();    #A

    public static String encode(byte[] data) {    #B
        return encoder.encodeToString(data);    #B
    }    #B

    public static byte[] decode(String encoded) {    #B
```